Database Design - Project Management Module

Overview

This document outlines the database schema for the Project Management Module based on the SRS requirements. The database will be implemented using MySQL with TypeORM.

Database Schema

1. Organization

Description: Top-level entity representing a company or organization
Table: `organization`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
name	VARCHAR(255)	NOT NULL	Organization name
settings	JSON	NULL	Organization-specific settings
plan	VARCHAR(50)	NULL	Subscription plan (e.g., 'free', 'pro', 'enterprise')
default_language	VARCHAR(10)	DEFAULT 'en'	Default language for organization (e.g., 'en', 'am', 'es')
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (name)

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2. User

Description: User accounts with roles and contact information
Table: `user`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
org_id	INT	FK, NOT NULL	Reference to organization
name	VARCHAR(255)	NOT NULL	Full name
email	VARCHAR(255)	UNIQUE, NOT NULL	Email address
password_hash	VARCHAR(255)	NOT NULL	Hashed password
role	ENUM	NOT NULL	Role: 'admin', 'manager', 'team_lead', 'reviewer', 'developer', 'executive'
specialty	VARCHAR(100)	NULL	Specialization (e.g., 'frontend', 'backend', 'ui_ux', 'ai')
phone	VARCHAR(20)	NULL	Phone number for SMS
telegram_id	VARCHAR(100)	NULL	Telegram user ID
github_username	VARCHAR(100)	NULL	GitHub username
github_token	VARCHAR(255)	NULL	Encrypted GitHub token
timezone	VARCHAR(50)	NULL	User's timezone
language	VARCHAR(10)	DEFAULT 'en'	User's preferred language (e.g., 'en', 'am', 'es')
status	ENUM	NOT NULL, DEFAULT 'active'	Status: 'active', 'inactive', 'suspended'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE INDEX (email)
- INDEX (org_id)
- INDEX (role)
- INDEX (status)
- INDEX (github_username)
- INDEX (telegram_id)

Foreign Keys:

- org_id → organization(id)

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3. Project

Description: Projects with goals, milestones, and deadlines
Table: `project`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
org_id	INT	FK, NOT NULL	Reference to organization
title	VARCHAR(255)	NOT NULL	Project title
description	TEXT	NULL	Detailed description
goals	TEXT	NULL	Project goals
tech_stack	JSON	NULL	Technology stack information
start_date	DATE	NULL	Project start date
deadline	DATE	NULL	Project deadline
update_interval_days	INT	DEFAULT 7	Progress update cycle in days
status	ENUM	NOT NULL, DEFAULT 'planning'	Status: 'planning', 'active', 'on_hold', 'completed', 'cancelled'
health	ENUM	DEFAULT 'on_track'	Health: 'on_track', 'at_risk', 'critical'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (org_id)
- INDEX (status)
- INDEX (health)
- INDEX (deadline)

Foreign Keys:

- org_id → organization(id)

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4. Milestone

Description: Project milestones with deadlines
Table: `milestone`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
project_id	INT	FK, NOT NULL	Reference to project
title	VARCHAR(255)	NOT NULL	Milestone title
description	TEXT	NULL	Milestone description
deadline	DATE	NOT NULL	Milestone deadline
status	ENUM	NOT NULL, DEFAULT 'pending'	Status: 'pending', 'in_progress', 'completed', 'delayed'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (project_id)
- INDEX (status)
- INDEX (deadline)

Foreign Keys:

- project_id → project(id) ON DELETE CASCADE

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5. Team

Description: Teams within projects (Frontend, Backend, UI/UX, AI)
Table: `team`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
project_id	INT	FK, NOT NULL	Reference to project
name	VARCHAR(100)	NOT NULL	Team name
type	ENUM	NOT NULL	Type: 'frontend', 'backend', 'ui_ux', 'ai'
lead_id	INT	FK, NULL	Team lead reference
deadline	DATE	NULL	Team deadline
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (project_id)
- INDEX (type)
- INDEX (lead_id)

Foreign Keys:

- project_id → project(id) ON DELETE CASCADE
- lead_id → user(id)

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6. TeamMember

Description: Many-to-many relationship between teams and users
Table: `team_member`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
team_id	INT	FK, NOT NULL	Reference to team
user_id	INT	FK, NOT NULL	Reference to user
role	VARCHAR(50)	NOT NULL	Role within team (e.g., 'member', 'lead')
joined_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Join timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE INDEX (team_id, user_id)
- INDEX (user_id)

Foreign Keys:

- team_id → team(id) ON DELETE CASCADE
- user_id → user(id) ON DELETE CASCADE

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7. Skill

Description: Catalog of skills that can be assigned to users
Table: `skill`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
name	VARCHAR(100)	NOT NULL	Skill name
specialty	VARCHAR(50)	NULL	Specialty category (e.g., 'frontend', 'backend')
status	ENUM	NOT NULL, DEFAULT 'active'	Status: 'active', 'inactive'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE INDEX (name)
- INDEX (specialty)
- INDEX (status)

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8. UserSkill

Description: Skills that users have declared with proficiency levels
Table: `user_skill`

Column	Type	Constraints	Description

id	INT	PK, AUTO_INCREMENT	Unique identifier
user_id	INT	FK, NOT NULL	Reference to user
skill_id	INT	FK, NOT NULL	Reference to skill
proficiency	ENUM	NOT NULL	Proficiency: 'beginner', 'intermediate', 'advanced', 'expert'
years_experience	INT	NULL	Years of experience
confirmed_by	INT	FK, NULL	User who confirmed this skill
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE INDEX (user_id, skill_id)
- INDEX (skill_id)
- INDEX (proficiency)

Foreign Keys:

- user_id → user(id) ON DELETE CASCADE
- skill_id → skill(id)
- confirmed_by → user(id)

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9. TaskRequiredSkill

Description: Skills required for a task
Table: `task_required_skill`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
task_id	INT	FK, NOT NULL	Reference to task
skill_id	INT	FK, NOT NULL	Reference to skill
minimum_proficiency	ENUM	NOT NULL	Minimum proficiency required

Indexes:

- PRIMARY KEY (id)
- UNIQUE INDEX (task_id, skill_id)
- INDEX (skill_id)

Foreign Keys:

- task_id → task(id) ON DELETE CASCADE
- skill_id → skill(id)

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10. Task

Description: Individual tasks within projects
Table: `task`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
project_id	INT	FK, NOT NULL	Reference to project
team_id	INT	FK, NULL	Reference to team
assignee_id	INT	FK, NULL	Reference to assigned user
parent_id	INT	FK, NULL	Parent task for subtasks
title	VARCHAR(255)	NOT NULL	Task title
description	TEXT	NULL	Task description
acceptance_criteria	TEXT	NULL	Acceptance criteria
priority	ENUM	NOT NULL, DEFAULT 'medium'	Priority: 'low', 'medium', 'high', 'critical'
estimate_hours	DECIMAL(5,2)	NULL	Estimated hours
status	ENUM	NOT NULL, DEFAULT 'todo'	Status: 'todo', 'in_progress', 'in_review', 'done', 'blocked'
deadline	DATE	NULL	Task deadline
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (project_id)
- INDEX (team_id)
- INDEX (assignee_id)
- INDEX (parent_id)
- INDEX (status)
- INDEX (priority)
- INDEX (deadline)

Foreign Keys:

- project_id → project(id) ON DELETE CASCADE
- team_id → team(id) ON DELETE SET NULL
- assignee_id → user(id) ON DELETE SET NULL
- parent_id → task(id) ON DELETE SET NULL

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11. TaskDependency

Description: Dependencies between tasks
Table: `task_dependency`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
task_id	INT	FK, NOT NULL	Task that depends
depends_on_task_id	INT	FK, NOT NULL	Task that is depended upon
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE INDEX (task_id, depends_on_task_id)
- INDEX (depends_on_task_id)

Foreign Keys:

- task_id → task(id) ON DELETE CASCADE
- depends_on_task_id → task(id) ON DELETE CASCADE

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12. Repository

Description: Git repositories linked to projects
Table: `repository`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
project_id	INT	FK, NOT NULL	Reference to project
provider	VARCHAR(50)	NOT NULL	Git provider (e.g., 'github')
repo_name	VARCHAR(255)	NOT NULL	Repository name (e.g., 'owner/repo')
repo_url	VARCHAR(500)	NOT NULL	Repository URL
webhook_secret	VARCHAR(255)	NULL	Webhook secret for verification
last_synced_at	TIMESTAMP	NULL	Last sync timestamp
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (project_id)
- UNIQUE INDEX (provider, repo_name)

Foreign Keys:

- project_id → project(id) ON DELETE CASCADE

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13. GitActivity

Description: GitHub activity linked to tasks (commits, PRs, reviews)
Table: `git_activity`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
repo_id	INT	FK, NOT NULL	Reference to repository
user_id	INT	FK, NULL	Reference to user
task_id	INT	FK, NULL	Reference to task
type	ENUM	NOT NULL	Type: 'commit', 'pull_request', 'review'
reference	VARCHAR(255)	NOT NULL	Reference (commit SHA, PR number, etc.)
status	VARCHAR(50)	NULL	Status (e.g., 'open', 'merged', 'closed')
metadata	JSON	NULL	Additional activity data
timestamp	TIMESTAMP	NOT NULL	Activity timestamp
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (repo_id)
- INDEX (user_id)
- INDEX (task_id)
- INDEX (type)
- INDEX (timestamp)

Foreign Keys:

- repo_id → repository(id) ON DELETE CASCADE
- user_id → user(id) ON DELETE SET NULL
- task_id → task(id) ON DELETE SET NULL

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14. Review

Description: Code reviews for pull requests or tasks
Table: `review`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
pr_or_task_id	INT	FK, NOT NULL	Reference to PR or task
reviewer_id	INT	FK, NOT NULL	Reference to reviewer
decision	ENUM	NULL	Decision: 'approved', 'changes_requested', 'rejected'
comment	TEXT	NULL	Review comments
review_date	TIMESTAMP	NULL	Review timestamp
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (pr_or_task_id)
- INDEX (reviewer_id)
- INDEX (decision)

Foreign Keys:

- reviewer_id → user(id)

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15. ProgressUpdate

Description: Developer progress updates
Table: `progress_update`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
user_id	INT	FK, NOT NULL	Reference to user
project_id	INT	FK, NOT NULL	Reference to project
cycle	INT	NOT NULL	Update cycle number
work_done	TEXT	NULL	Work completed
next_steps	TEXT	NULL	Planned next steps
blockers	TEXT	NULL	Any blockers reported

confidence	ENUM	NOT NULL	Confidence level: 'low', 'medium', 'high'
update_date	TIMESTAMP	NOT NULL	Update timestamp
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (user_id)
- INDEX (project_id)
- INDEX (cycle)
- INDEX (update_date)

Foreign Keys:

- user_id → user(id) ON DELETE CASCADE
- project_id → project(id) ON DELETE CASCADE

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16. Notification

Description: Notifications sent to users
Table: `notification`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
user_id	INT	FK, NOT NULL	Reference to user
type	VARCHAR(50)	NOT NULL	Notification type
channel	ENUM	NOT NULL	Channel: 'telegram', 'sms', 'email', 'in_app'
template	VARCHAR(100)	NULL	Template used
content	TEXT	NULL	Notification content
status	ENUM	NOT NULL, DEFAULT 'pending'	Status: 'pending', 'sent', 'delivered', 'failed'
attempts	INT	DEFAULT 0	Number of delivery attempts
sent_at	TIMESTAMP	NULL	Sent timestamp
delivered_at	TIMESTAMP	NULL	Delivered timestamp
error_message	TEXT	NULL	Error message if failed
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (user_id)
- INDEX (status)
- INDEX (channel)
- INDEX (created_at)

Foreign Keys:

- user_id → user(id) ON DELETE CASCADE

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17. EscalationRule

Description: Escalation rules for missed deadlines or updates
Table: `escalation_rule`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
org_id	INT	FK, NOT NULL	Reference to organization
scope	VARCHAR(50)	NOT NULL	Scope: 'project', 'task', 'update'
trigger	VARCHAR(100)	NOT NULL	Trigger condition
offset_hours	INT	NOT NULL	Hours after trigger to escalate
target_role	VARCHAR(50)	NULL	Target role for escalation
channel	ENUM	NOT NULL	Channel: 'telegram', 'sms', 'email'
is_active	BOOLEAN	DEFAULT TRUE	Whether rule is active
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (org_id)
- INDEX (scope)
- INDEX (is_active)

Foreign Keys:

- org_id → organization(id) ON DELETE CASCADE

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18. Policy

Description: Organizational policies
Table: `policy`

Column	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique identifier
org_id	INT	FK, NOT NULL	Reference to organization
name	VARCHAR(100)	NOT NULL	Policy name
description	TEXT	NULL	Policy description
config	JSON	NOT NULL	Policy configuration
is_active	BOOLEAN	DEFAULT TRUE	Whether policy is active
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last update timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (org_id)
- UNIQUE INDEX (org_id, name)
- INDEX (is_active)

Foreign Keys:

- org_id → organization(id) ON DELETE CASCADE

19. AuditLog

Description: Audit trail for all actions
Table: `audit_log`

Column	Type	Constraints	Description
id	BIGINT	PK, AUTO_INCREMENT	Unique identifier
org_id	INT	FK, NOT NULL	Reference to organization
actor_id	INT	FK, NOT NULL	Reference to user who performed action
action	VARCHAR(100)	NOT NULL	Action performed
entity_type	VARCHAR(50)	NOT NULL	Type of entity affected
entity_id	INT	NOT NULL	ID of entity affected
before_value	JSON	NULL	Value before change
after_value	JSON	NULL	Value after change
ip_address	VARCHAR(45)	NULL	IP address of actor
timestamp	TIMESTAMP	NOT NULL	Action timestamp

Indexes:

- PRIMARY KEY (id)
- INDEX (org_id)
- INDEX (actor_id)
- INDEX (entity_type, entity_id)
- INDEX (timestamp)

Foreign Keys:

- org_id → organization(id) ON DELETE CASCADE
- actor_id → user(id) ON DELETE SET NULL

Entity Relationship Diagram (ERD)

```
organization (1) ----< (N) user
organization (1) ----< (N) project
organization (1) ----< (N) escalation_rule
organization (1) ----< (N) policy
organization (1) ----< (N) audit_log

user (1) ----< (N) team_member
user (1) ----< (N) user_skill
user (1) ----< (N) task (as assignee)
user (1) ----< (N) git_activity
user (1) ----< (N) review (as reviewer)
user (1) ----< (N) progress_update
user (1) ----< (N) notification
user (1) ----< (N) audit_log (as actor)
user (1) ----< (N) team (as lead)

project (1) ----< (N) milestone
project (1) ----< (N) team
project (1) ----< (N) task
project (1) ----< (N) repository
project (1) ----< (N) progress_update

team (1) ----< (N) team_member
team (1) ----< (N) task

skill (1) ----< (N) user_skill
skill (1) ----< (N) task_required_skill

task (1) ----< (N) task_dependency (self-referencing)
task (1) ----< (N) task_required_skill
task (1) ----< (N) git_activity
task (1) ----< (N) review

repository (1) ----< (N) git_activity
```

Database Constraints and Rules

Referential Integrity

- All foreign key relationships enforce referential integrity
- ON DELETE CASCADE used for dependent entities
- ON DELETE SET NULL used for optional relationships

Data Validation

- ENUM types restrict values to valid options
- NOT NULL constraints ensure required fields
- UNIQUE constraints prevent duplicates
- CHECK constraints can be added for custom validation

Performance Considerations

- Indexes on foreign keys for faster joins
- Indexes on frequently queried columns (status, deadline, etc.)
- Composite indexes for common query patterns
- Timestamps for time-based queries

Security

- Passwords stored as hashes (bcrypt)
- GitHub tokens encrypted at rest
- Audit log captures all changes
- Role-based access enforced at application level

Data Retention

Based on the SRS requirements:

- Activity data: Configurable retention period (default: 1 year)

- Notifications: Configurable retention period (default: 6 months)
- Audit logs: Configurable retention period (default: 2 years)
- Personal data: Can be exported, anonymized, or deleted on request

Backup Strategy

- Daily full backups
- Continuous incremental backups
- Backups tested periodically
- Recovery Point Objective (RPO): 15 minutes
- Recovery Time Objective (RTO): 4 hours

Next Steps

1. Review and approve this database design
2. Generate TypeORM entity files
3. Create database migration scripts
4. Set up the MySQL database
5. Implement backend API with Express.js